

This **Level 2: Intermediate Level** resource is designed as a modular, step-by-step practical guide for the **Agri_Empower** digital library. It focuses on the transition from "growing" to "farming as a business."



LEVEL 2: INTERMEDIATE LEVEL

Goal: Professionalizing Farm Operations and Management Systems.

MODULE 1: ADVANCED PRODUCTION TECHNIQUES

Focus: Precision in field establishment and resource management.

- **Site Selection:** Not all land is equal. Choose sites with a gentle slope (for drainage), proximity to a reliable water source, and history check (avoid land where peppers or tomatoes grew last season to prevent soil-borne diseases).
- **Soil Testing Basics:** Use a pH meter or kit. Tomatoes thrive in **pH 6.0 to 6.8**. If the soil is too acidic, apply agricultural lime; if too alkaline, use organic matter or sulfur.
- **Raised Beds vs. Ridges:** * **Raised Beds:** Best for drip irrigation; they prevent soil compaction and improve aeration.
 - **Ridges:** Common for flood or bucket irrigation to channel water away from the plant base.
- **Fertilizer Scheduling:** Move from "random application" to a schedule.
 - *Establishment:* High Phosphorus (N-P-K 15-15-15) for root development.
 - *Vegetative:* Nitrogen for foliage.
 - *Fruiting:* High Potassium (K) for fruit size, color, and shelf life.
- **Irrigation Systems:**
 - **Drip:** The gold standard. Delivers water directly to roots, reducing weed growth and water waste.
 - **Bucket Irrigation:** Precise but labor-intensive; involves manual application at the plant base.
- **Crop Calendar Planning:** Working backward from your target harvest date. If market prices peak in July, count back 90–100 days to determine your nursery start date.



Module 1 Assessment

1. Calculate the number of stands for 1 hectare at 60cm x 40cm spacing.
2. Why is Potassium (K) critical during the fruiting stage specifically?

MODULE 2: ENHANCED BIOSECURITY & IPM

Focus: Shifting from "reactive spraying" to "proactive management."

- **Integrated Pest Management (IPM):** Combining physical (traps), biological (beneficial insects), and chemical (last resort) controls.
- **Pest Scouting Schedule:** Implement a 48-hour scouting rotation. Walk the farm in a "Z" pattern, checking the underside of leaves for eggs or larvae.
- **Spray Program Design:** Don't use the same chemical every week. Rotate **Active Ingredients** to prevent pests from developing resistance.
 - *Week 1:* Systemic insecticide.
 - *Week 2:* Contact fungicide.
- **Biosecurity Checklist:**
 - Restricted entry for visitors.
 - Dedicated boots for the farm zone.
 - Tools disinfected with a 10% bleach solution after every use.

Module 2 Assessment

1. What is the difference between a systemic and a contact pesticide?
2. Design a 4-week rotation schedule using two different fungicides.

MODULE 3: POST-HARVEST LOGISTICS

Focus: Maintaining the "Cold Chain" and quality standards.

- **Harvest Handling Protocol:** Harvest into small buckets, then transfer to crates. Never stack fruit more than 3–4 layers deep to avoid internal bruising.
- **Cold Chain Basics:** Field heat is the enemy. Move harvested tomatoes to a shaded, ventilated "holding area" immediately. Pre-cooling before transport extends shelf life by days.
- **Bulk Supply Prep:** For wholesalers, focus on uniformity in size. A single rotten tomato in a crate can trigger ethylene gas production, ripening and spoiling the rest prematurely.
- **Retail Packaging:** Moving toward "punnets" or shrink-wrapped trays. This protects the fruit and allows for branding and higher pricing in supermarkets.

Module 3 Assessment

1. How does "field heat" affect the shelf life of a tomato?
2. List three criteria for "Grade A" retail tomatoes.

MODULE 4: INTERMEDIATE PROCESSING

Focus: Small-scale industrialization for shelf-life extension.

- **Small-Scale Paste:** Involves washing, crushing, pulping, and boiling down the juice until it reaches a thick consistency (reducing moisture content to below 25%).
- **Dry Tomato Flakes:** Using a dehydrator or a solar dryer at **55°C–60°C**. Flakes can be ground into powder or stored as-is for soups and stews.
- **Shelf-Life Extension:** Use of acidity (lemon juice/vinegar) and heat-sealing in glass jars (canning) to keep sauces fresh for 6–12 months without refrigeration.
- **Packaging & Labeling:** Every product needs a Batch Number, Expiry Date, and NAFDAC-compliant ingredient list for commercial viability.

Module 4 Assessment

1. What is the primary purpose of reducing water content in tomato processing?
2. List the essential information required on a commercial food label.

MODULE 5: AGRIBUSINESS STRATEGY

Focus: Financial modeling and market dominance.

- **Budget Preparation:** Distinguish between **CAPEX** (Fixed costs like pumps, crates) and **OPEX** (Variable costs like seeds, fuel, labor).
- **Breakeven Analysis:** Knowing exactly how many kilograms you must sell to cover your total costs.
 - *Formula:* Total Cost / Average Selling Price = Breakeven Quantity.
- **Lean vs. Glut Season Strategy:** * *Glut (Dry Season):* Prices are low. Focus on processing and value addition.
 - *Lean (Rainy Season):* Prices are high. Focus on high-input production with rain shelters.
- **Institutional Supply:** Supplying hotels, restaurants, and schools. This requires consistency in quality and a formal invoice/receipt system.

Module 5 Assessment

1. If your total cost is ₦500,000 and you sell at ₦500/kg, how many kg do you need to sell to break even?

2. Identify one institutional buyer in your local area and list their likely quality requirements.
-



FINAL COURSE ASSIGNMENT

Task: Prepare a simple one-cycle tomato farm budget for a small plot (e.g., 1,000 stands). Include:

1. **Input Costs** (Seeds, Fertilizer, Chemicals).
 2. **Labor Costs** (Land prep, Weeding, Harvesting).
 3. **Logistics** (Crates, Transport).
 4. **Expected Revenue** (Projected yield x Estimated market price).
-

Administration Info

- **Contact:** 2348026279249 | agriempower4dcentury@gmail.com
- **Video Support:** Refer to the "Tomato Field Spacing" blueprint video on the Agri_Empower YouTube channel for practical visual layout guidance.